

MTH101 Midterm Past Paper File:02
Youtube:Life Star Academy
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Question No: 1 (Marks: 1) - Please choose one

If f is a twice differentiable function at a stationary point x_0 and $f''(x_0) > 0$ then f has relative At x_0

Minima

Maxima

None of these

Question No: 2 (Marks: 1) - Please choose one

If f is a twice differentiable function at a stationary point x_0 and $f''(x_0) < 0$ then f has relative At x_0

Minima

Maxima

None of these

Question No: 3 (Marks: 1) - Please choose one

A line $y = y_0$ is called a for the graph f if $\lim_{x \rightarrow +\infty} f(x) = y_0$ or $\lim_{x \rightarrow -\infty} f(x) = y_0$

Vertical asymptotes

Horizontal asymptotes

None of these

Question No: 4 (Marks: 1) - Please choose one

If $f(x) = 3x^8 + 2x + 1$ then $f'(x) =$ _____

$3x^7 + 2$

$24x^7 + 2$

$3x^9 + 2x^2$

$24x^9 + 2x^2$

Question No: 5 (Marks: 1) - Please choose one

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$$\frac{d(\tan x)}{dx} =$$

$$\sec x$$

$$\sec^2 x$$

$$\operatorname{cosec} x$$

$$\operatorname{cosec}^2 x$$

Question No: 6 (Marks: 1) - Please choose one

$$\text{If } xy = 4 \text{ then } \frac{dy}{dx} =$$

$$0$$

$$-\frac{1}{x^2}$$

$$\frac{4}{x^2}$$

$$-\frac{4}{x^2}$$

Question No: 7 (Marks: 1) - Please choose one

$$\text{If } 2x - y = -3 \text{ then } \frac{dy}{dx} =$$

$$2$$

$$-2$$

$$0$$

$$-3$$

Question No: 8 (Marks: 1) - Please choose one

$$\frac{d}{dx}[\sec x] = \underline{\hspace{2cm}}$$

$$\frac{1}{1 + \sin^2 x}$$

$$\frac{-\sin x}{1 + \sin^2 x}$$

$$\frac{1}{1 - \sin^2 x}$$

$$\frac{\sin x}{1 - \sin^2 x}$$

Question No: 9 (Marks: 1) - Please choose one

$$30^\circ = \underline{\hspace{2cm}}$$

$$\frac{\pi}{3}$$

$$\frac{\pi}{4}$$

$$\frac{\pi}{6}$$

$$\frac{\pi}{2}$$

Question No: 10 (Marks: 1) - Please choose one

Consider a function $h(x)$ and a constant c then

$$\frac{d}{dx}((c) \{h(x)\}) = \underline{\hspace{2cm}}$$

$$0$$

$$\frac{d}{dx}(h(x))$$

$$\frac{d}{dx}(h(cx))$$

$$c \frac{d}{dx}(h(x))$$

Question No: 11 (Marks: 1) - Please choose one

Suppose that f and g are differentiable functions of x then

$$\frac{d}{dx} \left[\frac{f}{g} \right] =$$

$$\frac{[g][f'] - [f][g']}{g^2}$$

$$\frac{[g'][f] - [f'][g]}{g^2}$$

$$\frac{[g][f']-[f][g']}{f^2}$$

$$\frac{[g'][f]-[f'][g]}{f^2}$$

Question No: 12 (Marks: 1) - Please choose one

$$\frac{d}{dx}[\operatorname{cosec} x] = \underline{\hspace{2cm}}$$

$$\frac{1}{1+\cos^2 x}$$

$$\frac{-\cos x}{1-\cos^2 x}$$

$$\frac{-\cos x}{1-\cos^2 x}$$

$$\frac{1}{1-\cos^2 x}$$

Question No: 13 (Marks: 1) - Please choose one

$$y = f(g(h(x)))$$

If

$$u = g(h(x))$$

$$v = h(x) \quad \text{then} \quad \frac{dy}{dx} = \underline{\hspace{2cm}}$$

$$\frac{dy}{du} \cdot \frac{du}{dv} \cdot \frac{dv}{dx}$$

$$\frac{dy}{du} \cdot \frac{du}{dv} \cdot \frac{dv}{dx}$$

$$\frac{dv}{du} \cdot \frac{du}{dv} \cdot \frac{dy}{dx}$$

Question No: 14 (Marks: 1) - Please choose one

Chain rule is a rule for differentiating _____ of functions.

Product

Sum

Difference
Composition

Question No: 15 (Marks: 1) - Please choose one

$$\frac{d}{dx}[x^n] = nx^{n-1}$$

The power rule, _____ holds if n is _____

- An integer
- A rational number
- An irrational number
- All of the above

Question No: 16 (Marks: 1) - Please choose one

Let a function f be defined on an interval, and let x_1 and x_2 denotes two distinct points in that interval. If $f(x_1) = f(x_2)$ for all points x_1 and x_2 then which of the following statement is correct?

- f is a decreasing function
- f is an increasing function
- f is a constant function

Question No: 17 (Marks: 1) - Please choose one

If $f''(x) < 0$ on an open interval (a, b) then which of the following statement is correct?

- f is concave up on (a, b) .
- f is concave down on (a, b)
- f is linear on (a, b) .

Question No: 18 (Marks: 1) - Please choose one

If $x > \frac{1}{2}$ then $\frac{d}{dx}[\ln 2x] =$ _____

- $\frac{1}{x}$
- $\frac{2}{x}$
- $\frac{1}{2x}$
- $\frac{2}{x^2}$

Question No: 19 (Marks: 1) - Please choose one

Let $y = (x^3 + 2x)^{37}$. Which of the following is correct?

$$\frac{dy}{dx} = (37)(x^3 + 2x)^{36}$$

$$\frac{dy}{dx} = 111x^2(x^3 + 2x)^{36}$$

$$\frac{dy}{dx} = (111x^2 + 74)(x^3 + 2x)^{36}$$

$$\frac{dy}{dx} = (111x^2 + 74)(x^3 + 2x)^{38}$$

Question No: 20 (Marks: 1) - Please choose one

If we have $x^2 + y^2 = 1$ then $\frac{dy}{dx} =$ _____

$$\frac{-x}{y}$$

$$\frac{x}{y}$$

$$\frac{-y}{x}$$

None of these

Question No: 21 (Marks: 1) - Please choose one

$$\log_b ac =$$

$$\log_b a + \log_b c$$

$$\log_b a - \log_b c$$

$$\frac{\log_b a}{\log_b c}$$

$$(\log_b a)(\log_b c)$$

Question No: 22 (Marks: 1) - Please choose one

$$\log_b a^r =$$

$$a \log_b r$$

$$r \log_b a$$

$$\frac{\log_b a}{\log_b r}$$

$$\log_b a + \log_b r$$

Question No: 23 (Marks: 1) - Please choose one

$$\log_b \frac{1}{c} = \underline{\hspace{2cm}}$$

$$\log_b c$$

$$1 - \log_b c$$

$$-\log_b c$$

$$1 + \log_b c$$

Question No: 24 (Marks: 1) - Please choose one

$$\log_b \frac{1}{t} = \underline{\hspace{2cm}}$$

$$\log_b t$$

$$1 - \log_b t$$

$$1 + \log_b t$$

$$-\log_b t$$

Question No: 25 (Marks: 3)

Differentiate:

$$f(t) = (t^3 + 4)^4$$

$$f'(t) = 4(t^3 + 4)^3 \cdot \frac{d}{dx}(t^3 + 4)$$

$$f'(t) = 4(t^3 + 4)^3 \cdot 3t^2$$

$$f'(t) = 12t^2(t^3 + 4)^3$$

Question No: 26 (Marks: 5)

$$\frac{\sqrt{13x^2 - 5x + 8}}{2\sqrt{13x^2 - 5x + 8}}$$

Differentiate $f(x) = \frac{1}{2\sqrt{13x^2 - 5x + 8}} \frac{d}{dx} 13x^2 - 5x + 8$

$$f'(x) = \frac{1}{2\sqrt{13x^2 - 5x + 8}} \cdot 26x - 5$$

Question No: 27 (Marks: 10)

Differentiate the following function

$$f(x) = x^3 \cdot e^{\frac{1}{x}}$$

$$f'(x) = 3x^2 \cdot e^{\frac{1}{x}} + x^3 \cdot e^{\frac{1}{x}} \cdot \left(-\frac{1}{x^2}\right)$$

$$f'(x) = 3x^2 \cdot e^{\frac{1}{x}} - \frac{x^3 \cdot e^{\frac{1}{x}}}{x^2}$$

$$f'(x) = e^{\frac{1}{x}} \left[3x^2 - \frac{x^3}{x^2} \right]$$

$$f'(x) = e^{\frac{1}{x}} \left[\frac{3x^4 - x^3}{x^2} \right]$$

$$f'(x) = x e^{\frac{1}{x}} \left[\frac{3x^3}{x^2} - \frac{x^2}{x^2} \right]$$

$$f'(x) = x e^{\frac{1}{x}} [3x - 1] \text{Ans}$$

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